BEING QUASI-FINITE, RESPECTIVELY LOCALLY QUASI-FINITE, IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Local finiteness of type and quasi-compactness descend along an fpqc covering. Same chapter, Lemmas lemma-descending-property-locally-finite-type and lemma-descending-property-quasi-compact, invoked at proof lines 6346-6349 and 6352-6354. Supply the two ingredients that together with local quasi-finiteness give quasi-finiteness.
- The base-change characterisation of locally quasi-finite morphisms. Morphisms, Lemma morphisms-lemma-base-change-quasi-finite, invoked at proof lines 6350-6352. Turns local finiteness of type plus the covering condition into local quasi-finiteness.

Source reference. Original lemma 02KX: sources/stacks-descent/pinned-original-source.tex, source lines 5954–5954. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TM: sources/stacks-chow/tags.txt, source lines 2379–2379. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The properties $\mathcal{P}(f)$ ="f is locally quasi-finite" and $\mathcal{P}(f)$ ="f is quasi-finite" are fpqc local on the base.*

Proof. Let $f : X \to S$ be a morphism of schemes, and let $\{S_i \to S\}$ be an fpqc covering such that each base change $f_i : X_i \to S_i$ is locally quasi-finite. We have already seen (Lemma 02KX) that "locally of finite type" is fpqc local on the base,

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and hence we see that f is locally of finite type. Then it follows from Morphisms, Lemma 01TM that f is locally quasi-finite. The quasi-finite case follows as we have already seen that "quasi-compact" is fpqc local on the base (Lemma 02KQ). $\square$